

Recommendation System

* Indicates required question

1. Email *

2. Roll Number *

3. Password *

Question

4. We have the following user–item rating matrix (blank = missing):

1 point

U1: I1 = 5, I2 = 3, I3 = –

U2: I1 = 4, I2 = –, I3 = 2

U3: I1 = –, I2 = 1, I3 = 2

We use a matrix–factorization model with biases:

$$r_{u,i} = \mu + b_u + b_i + p_u^T q_i$$

where

- μ is the global mean rating
- b_u is the user bias
- b_i is the item bias
- p_u, q_i are 2-dimensional latent vectors

Latent vectors are given as:

$p_{U1}=(1,2)$, $p_{U2}=(0,1)$, $p_{U3}=(2,1)$

$q_{I1}=(1,0)$, $q_{I2}=(0,1)$, $q_{I3}=(1,1)$

Biases are estimated by simple mean deviation.

Using this model, consider the missing rating for **user U2 on item I2**.

Check all that apply.

- ☐ Global mean rating μ is approximately 2.83.
- ☐ The predicted rating for U2 on I2, is approximately 3.17
- ☐ The user bias for U2 is approximately –0.17
- ☐ The item bias for I2 is approximately –0.33

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